# UNIT – IV

## Impact of Jets & Hydraulic Turbines

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

Syllabus: Hydrodynamic force of jets on stationary & moving flat, inclined and curved vanes, jet striking centrally and at tip, velocity diagrams, work done & efficiency, radial vanes. Hydraulic turbines – classification, impulse & reaction, Pelton, Francis, Kaplan, working proportions, work done, efficiencies, hydraulic design, draft tube.

## 1. Impact of Jets – Basics

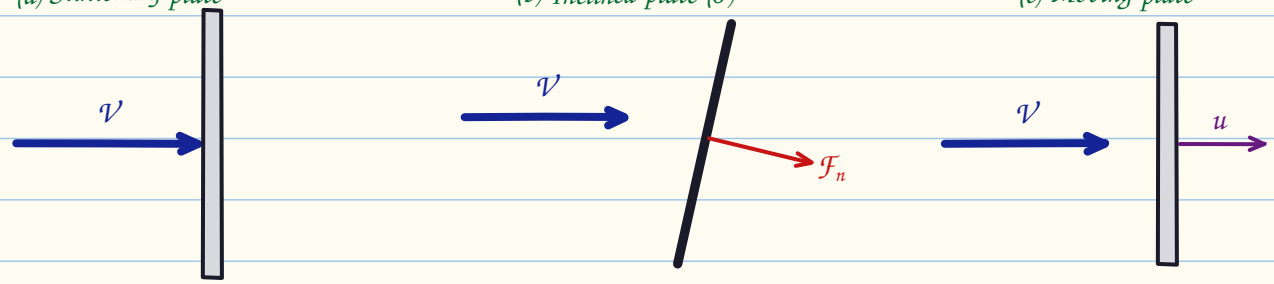
A jet of water leaving a nozzle at atmospheric pressure has only kinetic energy. When it strikes a plate or vane its momentum changes and a force is exerted. From the momentum equation :

$$F = \rho\, Q\,(V_{in} - V_{out}) \text{ (in the direction considered)}, \quad Q = a\,V$$

- $a$ = cross-section area of jet, $V$ = jet velocity, $u$ = velocity of the vane in the direction of the jet.
- Mass of water striking per second : stationary vane → $\rho a V$ ; single moving vane → $\rho a(V - u)$ ; series of vanes (wheel) → $\rho a V$ (all the water issuing from the nozzle is used because a new vane always takes the place of the old one).
- Force on the plate = rate of change of momentum ; Work done = $F \times u$ ; Jet power = $\frac{1}{2}\, \rho\, a\, V^3$ ; Efficiency $\eta$ = work done/ jet power.

## (A) Flat plates

(a) Stationary plate (b) Inclined plate (θ) (c) Moving plate




$F = \rho\, a\, V^2$ $F_n = \rho\, a\, V^2 \sin\theta$ $F = \rho\, a\, (V - u)^2$

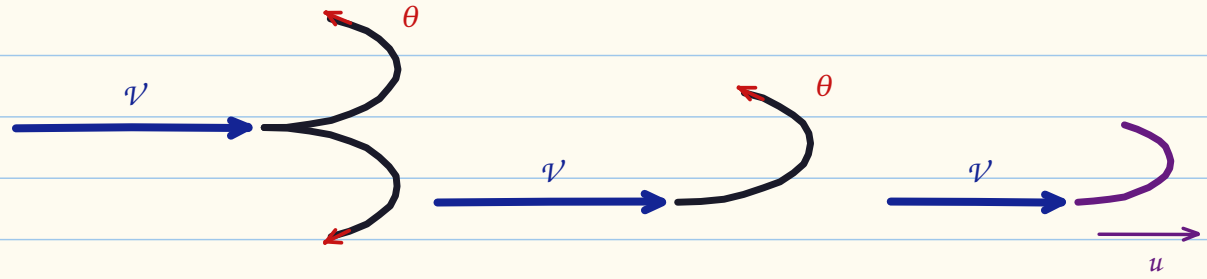
- ***Stationary plate, jet normal*** : $F = \rho\, a\, V^2$ (water leaves along the plate – no component in jet direction).
- ***Stationary inclined plate*** (plate inclined θ to jet axis, smooth) : normal force $F_n = \rho\, a\, V^2 \sin\theta$ ; component along jet $F_x = \rho\, a\, V^2 \sin^2\theta$ ; perpendicular to jet $F_y = \rho\, a\, V^2 \sin\theta \cos\theta$. The jet divides into two streams on the plate : $Q_1 = \frac{1}{2}Q(1 + \cos\theta)$ , $Q_2 = \frac{1}{2}Q(1 - \cos\theta)$.
- ***Hinged plate*** : plate hinged at top, jet strikes at its centre – equate moments : $F_n \times (L/2) = W \times (L/2) \sin\alpha \Rightarrow$ angle α of swing of the plate.
- ***Moving plate, normal*** (single) : $F = \rho\, a\, (V - u)^2$ ; work/sec $= F\, u$ ; $\eta = 2u(V - u)^2/V^3$ ; maximum $\eta = 8/27 = 29.6\ \%$ at $u = V/3$.
- ***Series of plates (wheel)*** : $F = \rho\, a\, V\, (V - u)$ ; work done $= \rho\, a\, V\, (V - u)\, u$ ;

$$\eta = 2u(V - u)/V^2 \quad \eta_{max} = 50\ \%\ \text{at}\ u = V/2$$

*Ex 1. Jet 50 mm dia, V = 20 m/s strikes a stationary flat plate normal : $F = 1000 \times (\pi/4)(0.05)^2 \times 20^2 = 1000 \times 0.0019635 \times 400 = 785$ N. If the plate moves at 5 m/s along the jet : $F = 1000(0.0019635)(15)^2 = 442$ N, work $= 442 \times 5 = 2.21$ kW.*

## (B) Curved vanes

(a) Stationary, jet at centre (b) Jet at tip (tangential) (c) Moving vane




$F = \rho a V^2(1 + \cos\theta)$ $F_x = \rho a V^2(1 + \cos\theta)$ $F_x = \rho a(V-u)^2(1+\cos\theta)$

Curved vanes turn the jet through a large angle, so the momentum change is greater. Let $\theta$ = angle between the jet leaving the tip and the direction of motion (outlet angle). A smooth vane has $V_{r2} = V_{r1}$.

- ***Stationary symmetrical vane, jet at centre*** : jet splits equally ; force in jet direction : $F_x = \rho\ a\ V^2(1 + \cos\theta)$. For a semi-circular vane ($\theta = 0$) $F = 2\rho a V^2$ – twice that on a flat plate.
- ***Stationary unsymmetrical vane, jet at one tip (tangential)*** : $F_x = \rho\ a\ V^2(\cos\theta + \cos\varphi)$ , $F_y = \rho\ a\ V^2(\sin\theta - \sin\varphi)$ where $\theta$ , $\varphi$ are inlet and outlet tip angles. Resultant $F = \sqrt{(F_x^2 + F_y^2)}$.
- ***Moving curved vane (single)*** : $F_x = \rho\ a\ (V - u)^2(1 + \cos\theta)$.
- ***Series of curved vanes (wheel, e.g. Pelton)*** : mass $\rho a V$ per second ; relative velocity at inlet $V_{r1} = V - u$ ; at outlet $V_{r2} = V_{r1}$ (smooth) ; whirl at outlet $V_{w2} = V_{r2}\cos\varphi - u$ ($\varphi$ = outlet vane angle with the –ve direction of u).

$$F_x = \rho\ a\ V\ (V - u)(1 + \cos\varphi)\quad Work = F_x\ u$$

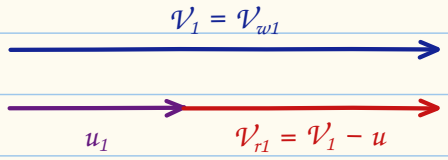
$$\eta = 2u\ (V - u)(1 + \cos\varphi)\ /\ V^2\quad \text{maximum at } u = V/2 : \eta_{max} = (1 + \cos\varphi)/2$$

- For $\varphi = 0$ (semicircular vane, ideal) $\eta_{max} = 100\ \%$ – all the KE is converted. In practice $\varphi \approx 15° \Rightarrow \eta_{max} \approx 98\ \%$.

Ex 2. Jet 75 mm dia, $V = 30$ m/s strikes the centre of a stationary symmetrical vane that deflects it through 150° ($\theta = 30°$). $F = \rho a V^2(1 + \cos 30°) = 1000 \times 0.004418 \times 900 \times 1.866 =$ 7420 N.

## Velocity diagrams – work done

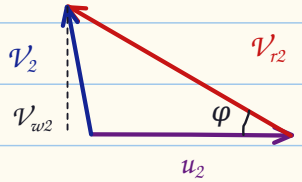
INLET triangle ($\alpha = 0$)




Pelton / moving vane

$V_{f1} = 0$

OUTLET triangle




$V_{w2} = V_{r2}\cos\varphi - u_2$

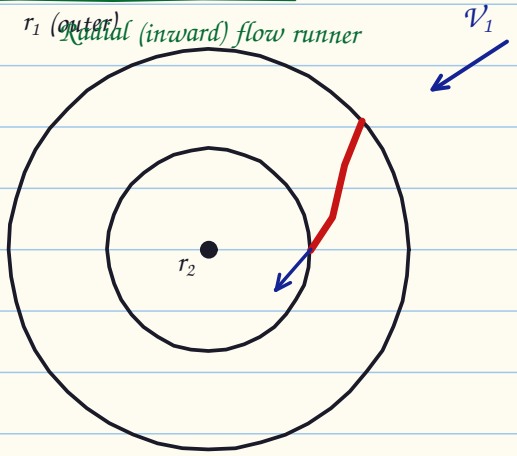
For a general moving vane (inlet angle $\alpha$, vane angles $\theta$ and $\varphi$) the force in the direction of motion is $F_x = \rho\, Q\,(V_{w1} \pm V_{w2})$ ; '+' when the outlet whirl $V_{w2}$ is in the opposite direction to $u$ (vane angle $\varphi < 90°$), '–' when in the same direction.

**Work done per second = $\rho\, Q\,(V_{w1} \pm V_{w2})\, u$ Work done per kg = $(V_{w1} \pm V_{w2})\, u$**

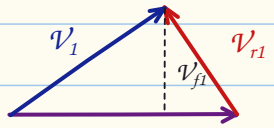
## Jet striking at the tip vs centrally

- Centre : symmetrical vane, jet splits, side thrust cancels – only axial force ; used in Pelton buckets (splitter at the centre).
- Tip : unsymmetrical vane, flow enters tangentially – there is side force $F_y$; ensures shock-free entry when vane angle = jet direction.

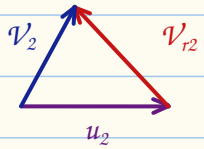
## Flow over radial vanes




INLET ($\alpha$ , $\theta$)




OUTLET ($\beta$ , $\varphi$)




$W = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$

In a wheel with radial vanes (inlet radius $r_1$, outlet $r_2$) the velocities are tangential ($u = \omega\, r$), so the angular momentum equation gives torque :

**$T = \rho\, Q\,(\, V_{w1}\, r_1 \pm V_{w2}\, r_2\,)$ Work done = $T\,\omega = \rho\, Q\,(V_{w1}u_1 \pm V_{w2}u_2)$**

- Radial inward flow (turbine) : water enters at outer radius $r_1$, leaves at inner $r_2$. Radial outward flow (centrifugal pump) is the reverse and the work is added to the fluid.
- Hydraulic efficiency $\eta_h = (V_{w1}u_1 \pm V_{w2}u_2) / (g\,H)$. This is the **Euler turbine equation** – basis of all turbomachines.

Ex 3. Radial vane wheel : $u_1 = 25$ m/s, $V_{w1} = 18$ m/s, $u_2 = 12$ m/s, $V_{w2} = 0$ (radial discharge), $Q = 0.5\ m^3/s \Rightarrow$ work $= 1000 \times 0.5 \times (18 \times 25) =$ 225 kW.

## 2. Hydraulic Turbines – Classification

Turbines convert the hydraulic energy of water (head) into mechanical energy of a rotating shaft, which drives a generator. Gross head $H_g$ = head race level – tail race level ; net (effective) head $H = H_g - h_f$ (friction loss in penstock).

| Basis | Types | Example |
|---|---|---|
| Energy at inlet | Impulse (only KE ; atmospheric) / Reaction (KE + pressure) | Pelton / Francis, Kaplan |
| Direction of flow | Tangential / radial (inward) / axial / mixed | Pelton / Francis (old) / Kaplan / modern Francis |
| Head | High (> 250 m) / medium (60–250 m) / low (< 60 m) | Pelton / Francis / Kaplan, propeller |
| Specific speed $N_s$ | Low (< 60) / medium (60–400) / high (> 400) | Pelton / Francis / Kaplan |
| Shaft position | Horizontal / vertical | Pelton often H ; Francis, Kaplan V |
| Discharge | Low / medium / high | Pelton / Francis / Kaplan |

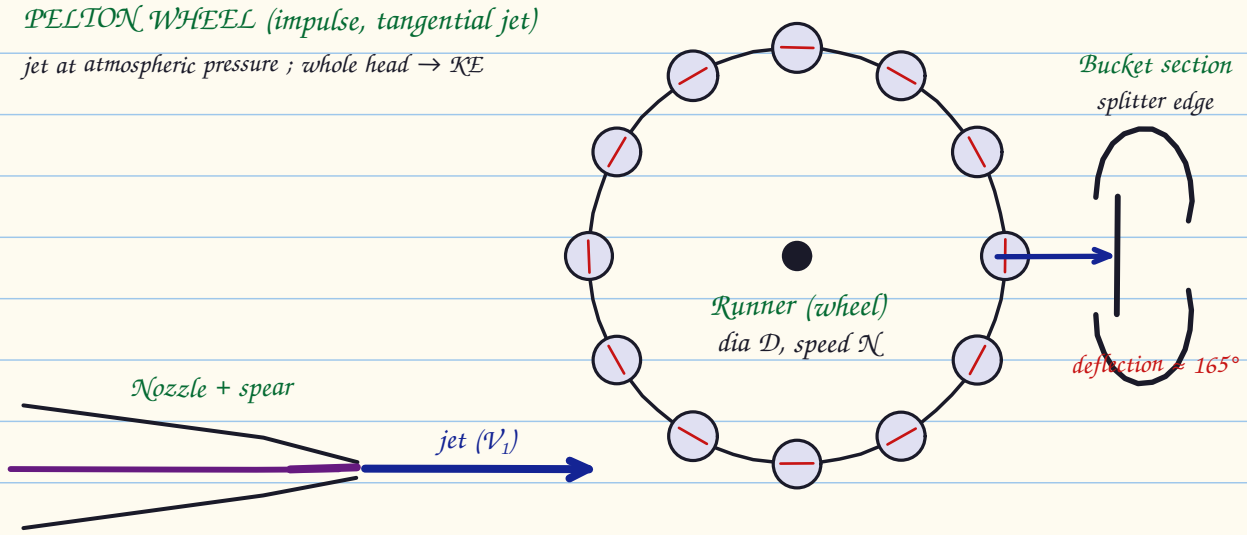
### Efficiencies & power

Water power $WP = \rho\, g\, Q\, H$ Runner power $RP = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$ Shaft power SP

| Efficiency | Definition | Typical value |
|---|---|---|
| Hydraulic $\eta_h$ | RP / WP = runner power / water power | 85 – 94 % |
| Mechanical $\eta_m$ | SP / RP (bearing, windage loss) | 95 – 99 % |

| Efficiency | Definition | Typical value |
|---|---|---|
| Overall $\eta_o$ | $SP / WP = \eta_h \times \eta_m$ | 80 – 92 % |
| Volumetric $\eta_v$ | $(Q - q_{leak})/Q$ | 97 – 99 % |

## 3. Pelton Wheel (Impulse Turbine)




**Working** : water from the penstock reaches the nozzle, whose spear controls the jet diameter. The whole head is converted to KE ; the jet at atmospheric pressure strikes the double-cup buckets at the splitter, is turned through about 165° and falls into the tail race. Runner rotates in air – casing only prevents splashing. A brake nozzle stops the wheel quickly.

### Parts

- **Nozzle and spear** (needle) – regulates discharge ; **deflector** – diverts the jet from the buckets on load rejection to avoid water hammer ; **buckets** – double hemispherical cups with a central splitter ; **casing** ; **penstock** ; **runner disc.**

### Velocities, work done and efficiency

- Jet velocity $V_1 = C_v \sqrt{(2gH)}$ ($C_v \approx 0.97$–$0.99$). Bucket velocity $u = \varphi_s \sqrt{(2gH)}$ where speed ratio $\varphi_s \approx 0.43 - 0.48$.
- $V_{w1} = V_1$ ; $V_{r1} = V_1 - u$ ; $V_{r2} = k\, V_{r1}$ ($k \approx 0.9$–$0.95$ for friction) ; $V_{w2} = V_{r2} \cos\varphi - u$ ; $\varphi = 180° -$ bucket angle $\approx 15°$.

$$\text{Work done / s} = \rho\, Q\, (V_1 - u)(1 + k\cos\varphi)\, u$$

$$\eta_h = 2\,(V_1 - u)(1 + k\cos\varphi)\, u / V_1^2 \quad \eta_{max} \text{ when } u = V_1/2 : (1 + k\cos\varphi)/2$$

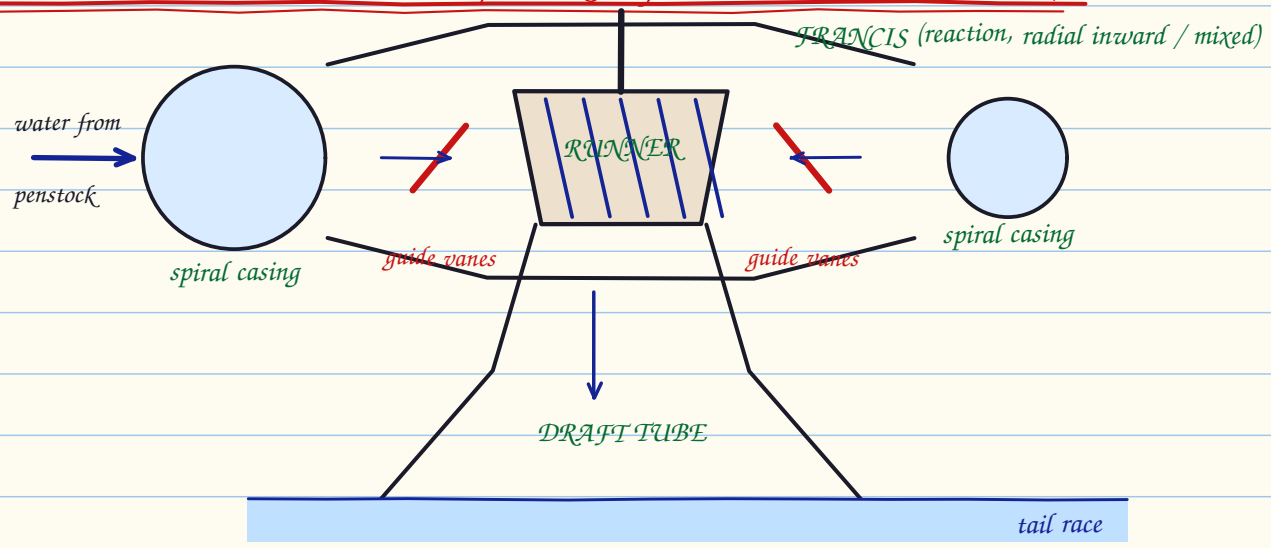
- Power available at nozzle = $\rho g Q H$ ; $N = 60u/(\pi D)$.

## Hydraulic design (working proportions)

- Jet diameter $d = \sqrt{(4Q/(\pi V_1))}$. **Jet ratio** $m = D/d = 10 – 20$ (12 normally) ; number of buckets $z = 15 + D/(2d) = 15 + m/2$.
- Bucket : width ≈ 5d, depth ≈ 1.2d ; deflection 165°. Maximum jets per wheel (vertical shaft) : 6. Specific speed 10 – 60 for one jet (increases as √(number of jets)).
- Governing : spear moves in/out to change Q while V stays the same ⇒ efficiency remains almost constant over wide load range.

Ex 4. Pelton wheel : $H = 300$ m, $D = 1.5$ m, jet $d = 100$ mm, $C_v = 0.98$, $\varphi_s = 0.45$, $k = 1$, $\varphi = 15°$. $V_1 = 0.98\sqrt{(2\times9.81\times300)} = 75.2$ m/s ; $Q = 0.007854 \times 75.2 = 0.59$ $m^3/s$ ; $u = 0.45 \times 76.7 = 34.5$ m/s ⇒ $N = 60u/\pi D =$ 440 rpm. $V_{r1} = 40.7$ ; $V_{w2} = 40.7\cos15° – 34.5 = 4.8$ m/s. Power = $1000 \times 0.59 \times (75.2 + 4.8) \times 34.5$ = 1630 kW ; WP = 1738 kW ⇒ $\eta_h =$ 93.8 %.

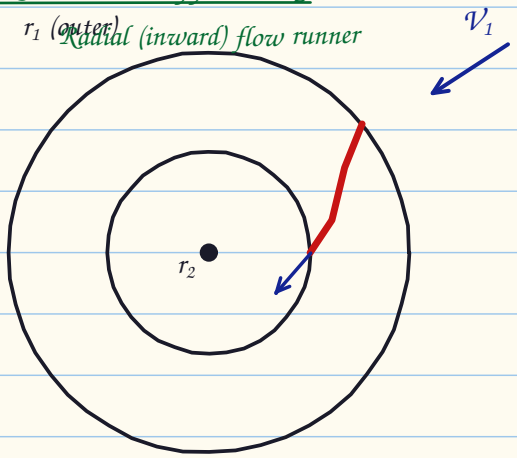
## 4. Francis Turbine (Radial / Mixed-flow Reaction Turbine)

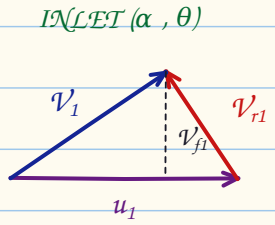


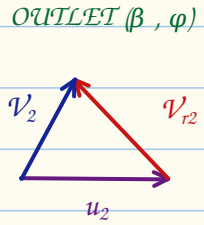

**Working** : water from the penstock enters the spiral (scroll) casing, whose cross-section decreases so that velocity stays uniform around the periphery. It passes through stay vanes and adjustable **guide vanes (wicket gates)** which give the flow the correct angle α, and enters the runner radially, flowing inward and leaving axially into the **draft tube.** The runner is completely filled with water, at pressure above atmosphere at inlet and below at exit – both pressure and kinetic energy are converted.

- Parts : spiral casing, stay ring, guide vanes (governor linkage), runner (16–24 blades, curved backward), draft tube, shaft.

## Work done & efficiency




$W = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$

$$\text{Work/s} = \rho\, Q\,(V_{w1}\,u_1 \pm V_{w2}\,u_2)\,\eta_h = (V_{w1}u_1 \pm V_{w2}u_2)\,/\,(g\,H)$$

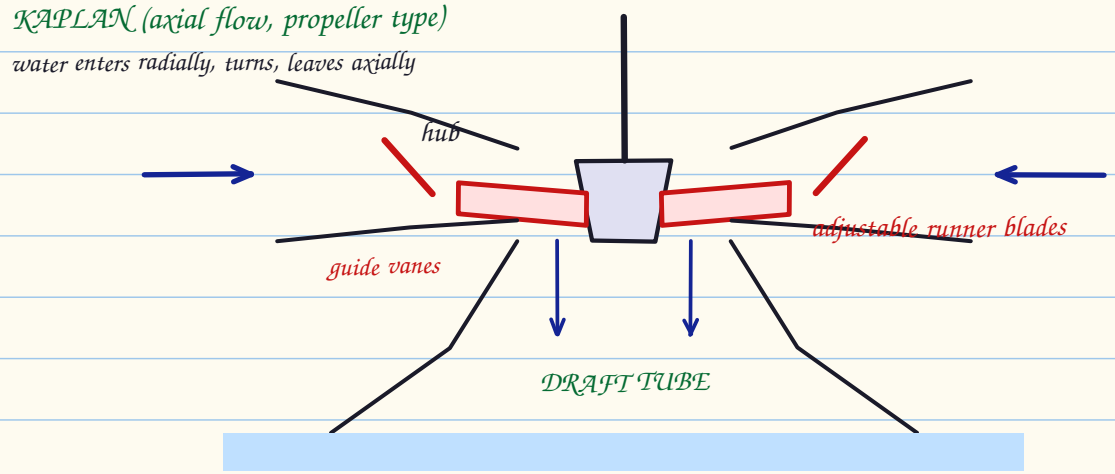
- For radial discharge ($V_{w2} = 0$, exit blade angle ~ 90°) : $\eta_h = V_{w1}\,u_1 / (g\,H)$. Discharge $Q = \pi\, D_1\, B_1\, V_{f1} = \pi\, D_2\, B_2\, V_{f2}$ (reduce by vane thickness factor $k_t \approx 0.95$).
- **Degree of reaction** $R$ = (pressure energy change in runner) / (total energy change) $= 1 - (V_1^2 - V_2^2)/(2\,g\,E)$ , Francis $R \approx 0.4$–$0.6$ ; impulse $R = 0$ ; Kaplan $R \approx 0.5$-$0.7$.

## Working proportions & design

- Speed ratio $K_u = u_1/\sqrt{(2gH)} = 0.6 - 0.9$ ; flow ratio $K_f = V_{f1}/\sqrt{(2gH)} = 0.15 - 0.30$. Breadth ratio $n = B_1/D_1 = 0.10 - 0.45$ ; ratio $D_1/D_2 \approx 1.5$-$2$ ; guide vane angle $\alpha = 15°$–$30°$ ; runner vane angle at inlet $\theta = 60°$–$120°$.
- Governing by changing the opening of the guide vanes ; $N = 60u_1/(\pi D_1)$.

Ex 5. Francis : $D_1 = 1.2$ m, $N = 300$ rpm, $B_1 = 0.15$ m, $\alpha = 20°$, $V_{f1} = 6$ m/s, $H = 40$ m, radial discharge. $u_1 = \pi(1.2)(300)/60 = 18.85$ m/s ; $V_{w1} = 6/\tan 20° = 16.5$ m/s ; $\eta_h = 16.5\times18.85/(9.81\times40) = 79\ \%$ ; $Q = \pi\, D_1\, B_1\, V_{f1} = 3.39\ m^3/s$ ; power $= 1000 \times 3.39 \times 16.5 \times 18.85 = 1.05\ MW$.

## 5. Kaplan Turbine (Axial-flow Reaction Turbine)




- A propeller turbine with adjustable runner blades (and adjustable guide vanes → double regulation). Water enters through the spiral casing and guide vanes, turns through 90° in the vaneless space and flows axially through the runner. Used for low head (5 – 60 m) and very large discharge. Propeller turbine = fixed blades (good only near design load).
- Parts : scroll casing, guide vanes, hub (boss) carrying 3 – 8 blades, draft tube. Blade angle is adjusted by a servo mechanism in the hub as load varies, keeping efficiency high from 30 % to 100 % load.
- Since flow is axial, $u_1 = u_2 = \pi D_o N/60$ (at blade outer rim) ; $D_o$ = runner dia, $D_b$ = boss dia.

$$Q = (\pi/4)(D_o^2 - D_b^2)\, V_{f1} \quad \text{Work/s} = \rho\, Q\,(V_{w1} \pm V_{w2})\, u \quad \eta_h = (V_{w1} \pm V_{w2})\, u/gH$$

- Proportions : hub ratio $D_b/D_o$ = 0.35 – 0.6 ; speed ratio 1.4 – 2.0 ; flow ratio 0.5 – 0.7 ; specific speed 300 – 1000.

Ex 6. Kaplan : $H$ = 20 m, $D_o$ = 3 m, $D_b$ = 1.2 m, $K_f$ = 0.5, $\eta_o$ = 0.88. $V_{f1} = 0.5\sqrt{(2\times9.81\times20)} = 9.9$ m/s ; $Q = (\pi/4)(9 - 1.44)(9.9) = 58.8$ m³/s ; $P = \eta_o \rho g Q H = 0.88 \times 9810 \times 58.8 \times 20 =$ 10.2 MW.

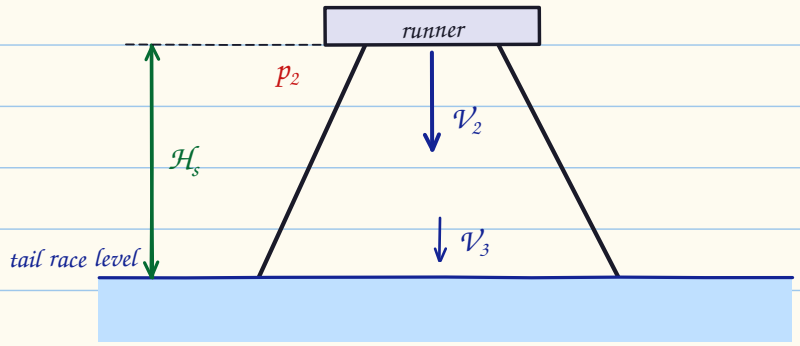
### Comparison of the three turbines

| Item | Pelton | Francis | Kaplan |
|---|---|---|---|
| Type | Impulse | Reaction | Reaction |
| Head | High > 250 m | Medium 60–250 m | Low < 60 m |
| Discharge | Low | Medium | High |
| Flow direction | Tangential | Radial → axial | Axial |

| Item | Pelton | Francis | Kaplan |
|---|---|---|---|
| $N_s$ | 10 – 60 | 60 – 400 | 300 – 1000 |
| Regulation | Spear + deflector | Guide vanes | Guide vanes + blades |
| Draft tube | Not needed | Needed | Needed |
| Efficiency curve | flat at part load | falls at part load | flat (double regulation) |

## 6. Draft Tube

A gradually expanding conduit connecting the runner exit to the tail race, completely submerged at its outlet (used in reaction turbines).




$\eta_d = [(V_2^2 - V_3^2)/2g - h_f] / (V_2^2/2g)$

$p_2/\rho g = p_{atm}/\rho g - H_s - \eta_d V_2^2/2g$

($p_2$ below atmospheric ⇒ extra head)

conical tube : divergence 7–8° , $\eta_d \approx 90$ %

### Functions

- Permits the turbine to be set above tail-water level (for easy inspection/maintenance) without losing the head $H_s$ – because the pressure at runner exit is below atmospheric (suction).
- Converts the large kinetic energy leaving the runner ($V_2^2/2g$) into pressure energy by gradual expansion – recovered as additional effective head ; reduces the exit loss to $V_3^2/2g$.
- Keeps the discharge from the runner under water seal and prevents air from entering.

### Theory & efficiency

Bernoulli between runner exit 2 and tail-race level 3 : $p_2/\rho g + V_2^2/2g + H_s = p_{atm}/\rho g + V_3^2/2g + h_f$.

$$\eta_d = [\,(V_2^2 - V_3^2)/2g - h_f\,] / (\,V_2^2/2g\,) \quad p_2/\rho g = p_{atm}/\rho g - H_s - \eta_d\, V_2^2/2g$$

- Larger $\eta_d$ ⇒ lower pressure at runner exit ⇒ more head utilised – but pressure must stay above the vapour pressure (cavitation limit – see Unit V).

| Type | Description | $\eta_d$ |
|---|---|---|
| Conical (straight divergent) | cone angle ≤ 8–10° ; $H_s$ < 8 m | ≈ 90 % |
| Simple elbow | elbow with constant / expanding section | 60 – 65 % |
| Moody spreading (hydracone) | conical with central bell | up to 85 % |
| Elbow with varying section | bend + rectangular outlet (most common for Francis/Kaplan) | ≈ 85 % |

- Divergence angle must be ≤ 10° – beyond this the flow separates from the wall (adverse pressure gradient) and the tube loses its recovery.

Ex 7. Draft tube inlet dia 1.0 m, outlet 1.6 m, $Q = 3\ m^3/s$ : $V_2 = 3.82$ , $V_3 = 1.49\ m/s$ ; if $h_f = 0.1\ m$ : $\eta_d = [(14.6 - 2.22)/19.62 - 0.1]/(14.6/19.62) = (0.631 - 0.1)/0.744 = 71\ \%$.

## 7. Quick Formula Sheet – Unit IV

| Item | Formula |
|---|---|
| Flat plate normal | $F = \rho a V^2$ ; moving: $\rho a(V-u)^2$ ; wheel: $\rho a V(V-u)$ |
| Inclined plate | $F_n = \rho a V^2 \sin\theta$ |
| Curved vane (wheel) | $F_x = \rho a V(V-u)(1+\cos\varphi)$ |
| Efficiency (wheel) | $2u(V-u)(1+\cos\varphi)/V^2$ ; max at $u = V/2$ |
| Euler equation | $W = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$ |
| Pelton | $V_1 = C_v\sqrt{(2gH)}$ ; $u = 0.45\sqrt{(2gH)}$ ; $m = D/d$ |
| Francis | $\eta_h = V_{w1}u_1/gH$ ; $Q = \pi D_1 B_1 V_{f1}$ |
| Kaplan | $Q = (\pi/4)(D_o^2 - D_b^2)V_f$ |
| Efficiencies | $\eta_o = \eta_h\eta_m = SP/WP$ |

## 8. Important Questions

- Derive the force exerted by a jet on a (i) stationary (ii) moving flat plate; find the condition for maximum efficiency of a series of plates.
- Obtain expressions for work done and efficiency of a jet on a series of curved vanes; show $\eta_{max} = 50\ \%$ for flat and $(1+\cos\varphi)/2$ for curved vanes.

- Draw velocity triangles for a Pelton bucket and derive efficiency; explain governing and working proportions.
- Classify hydraulic turbines. Define different efficiencies of a turbine.
- Explain with neat sketch the working of Francis turbine; derive the work done and efficiency; define degree of reaction.
- Explain Kaplan turbine and compare it with propeller and Francis turbines.
- What is a draft tube? Explain its functions, types and efficiency; derive the pressure at runner exit.